

Zomato Restaurant Dataset – Data Analysis Report

1. Dataset Overview

The analysis uses a Zomato restaurant dataset containing information about restaurants, locations, cuisines, pricing, online delivery, table booking, ratings and votes. The notebook initially loads 9,552 records with 22 columns. Key fields include Restaurant ID, Restaurant Name, City, Cuisines, Average Cost for two, Price range, Aggregate rating and Votes.

The dataset covers 141 unique cities. New Delhi has the largest restaurant presence in the dataset, with 5,473 restaurants, followed by Gurgaon (1,118) and Noida (1,080). North Indian is the most frequently occurring cuisine, appearing 936 times.

2. Data Cleaning

Missing-value inspection identified limited missing data. Locality and Locality Verbose each had 1 missing value; Longitude, Latitude, Average Cost for two, Currency, Has Table booking, Has Online delivery, Is delivering now and Switch to order menu each had 1 missing value. Cuisines had 10 missing values, while Price range, Aggregate rating, Rating color, Rating text and Votes each had 2 missing values.

For cleaning, the notebook filtered the Average Cost for two column to retain values that could be converted to numeric, then converted that column from object to numeric. The Country Code column was converted from float to object because it represents categorical country identifiers. Missing cuisine values were replaced with 'Unknown'. The duplicate-record check returned an empty DataFrame, indicating no exact duplicate rows were found.

After the numeric filtering step, the statistical analysis operates on 9,550 observations. The notebook does not document additional imputation for the remaining missing fields.

3. Data Analysis

The statistical summary for the cleaned analytical data shows an average cost for two of 1,199.33, an average price-range score of 1.80, an average aggregate rating of 2.67, and an average of 156.92 votes. The median cost is 400, the median price range is 2, the median rating is 3.2 and the median vote count is 31. The cost variable is highly dispersed, with a maximum of 800,000 compared with a median of 400, indicating substantial skew/outlier influence.

The notebook's analysis questions produced the following results:

Average restaurant rating: 2.6663.

Restaurant with the highest number of votes: Toit, with 10,934 votes.

Average cost for two: 1,199.33.

Average rating by online delivery: 2.47 for restaurants without online delivery and 3.25 for restaurants with online delivery.

Average rating by table booking: 2.56 for restaurants without table booking and 3.44 for restaurants with table booking.

Top 10 cities by restaurant count: New Delhi, Gurgaon, Noida, Faridabad, Ghaziabad, Bhubaneshwar, Lucknow, Ahmedabad, Amritsar and Guwahati.

4. Conclusion

The analysis provides a concise view of restaurant distribution, pricing, cuisines and customer ratings in the Zomato dataset. Restaurant activity is heavily concentrated in a small number of cities, especially New Delhi. North Indian cuisine is the most common single cuisine category. Restaurants with online delivery and table-booking facilities show higher average ratings than restaurants without those facilities.

From a business perspective, the results suggest that digital ordering and reservation capabilities are associated with stronger average ratings in this dataset. The concentration of restaurants in a few cities may also indicate highly competitive markets and opportunities to examine less-served locations. Finally, the large gap between the median and maximum average cost for two highlights the need to consider skew and outliers when using price-related statistics for business decisions.